

Dual Antiplatelet Therapy and Immediate Intensive Statin in Mild Ischemic Stroke

A Randomized Trial

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Abstract

Background and Objectives

Previous studies have shown a beneficial effect of clopidogrel-aspirin and intensive statin therapy in acute ischemic stroke; however, the synergistic effect of the 2 treatments is still unclear. The aim of this study was to investigate the effect of combining clopidogrel-aspirin and immediate intensive statin in patients with acute mild ischemic stroke or transient ischemic attack (TIA).

Methods

We performed a multicenter, randomized, double-blind, placebo-controlled trial with a 2-by-2 factorial design across 222 hospitals in China. Eligible participants were patients with acute mild ischemic stroke or TIA of a presumed atherosclerotic cause within 72 hours of symptom onset. Patients were randomly assigned to receive clopidogrel plus aspirin or aspirin alone and an immediate or delayed intensive statin. The primary efficacy outcome was a new stroke (ischemic or hemorrhagic) within 90 days, and the primary safety outcome was moderate-to-severe bleeding.

Results

Between September 17, 2018, and October 15, 2022, 6,100 patients were enrolled (median age, 65 years; 64.2% male), of whom 1,525 each were assigned to the 4 groups. New stroke within 90 days occurred in 116 patients (7.6%) in the clopidogrel-aspirin plus immediate intensive statin group (hazard ratio [HR] 0.76, 95% CI 0.60–0.97), in 106 patients (7.0%) in the clopidogrel-aspirin plus delayed statin group (HR 0.69, 95% CI 0.54–0.89), and in 129 patients (8.5%) in the aspirin plus immediate statin group (HR 0.85, 95% CI 0.67 to 1.07), compared with 150 patients (9.9%) in the aspirin plus delayed intensive statin group. Moderate-to-severe bleeding occurred in 17 (1.1%) in the clopidogrel-aspirin plus immediate statin group ($p = 0.047$), 10 (0.7%) in the clopidogrel-aspirin plus delayed statin group ($p = 0.46$), and 6 (0.4%) in the aspirin plus immediate statin group ($p = 0.80$), compared with 7 (0.5%) in the aspirin plus delayed statin group.

Discussion

Among patients with mild ischemic stroke or TIA of presumed atherosclerotic cause, the combination of clopidogrel-aspirin and delayed intensive statin was superior to aspirin plus delayed intensive statin in reducing the risk of new stroke, an effect that was mainly driven by clopidogrel-aspirin and not significantly different from that of clopidogrel-aspirin plus immediate statin. The combination treatment had a low but increased risk of moderate-to-severe bleeding.

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Class of Evidence

Criteria for rating therapeutic and diagnostic studies

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Supplementary Material

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Glossary

cOR = common odds ratio; **DAPT** = dual antiplatelet therapy; **HR** = hazard ratio; **INSPIRES** = Intensive Statin and Antiplatelet Therapy for Acute High-risk Intracranial or Extracranial Atherosclerosis; **mRS** = modified Rankin Scale; **NIHSS** = NIH Stroke Scale.

Trial Registration Information

ClinicalTrials.gov identifier: NCT03635749.

Classification of Evidence

This study provides Class I evidence that in patients with mild ischemic stroke or TIA of presumed atherosclerotic cause, the combination of clopidogrel-aspirin plus delayed intensive statin was superior to aspirin plus delayed intensive statin in reducing the 90-day risk of new stroke.

Introduction

Patients with acute ischemic stroke or TIA are at high risk of new stroke and poor functional outcome within 90 days^{1,2}. Antiplatelet therapy and intensive statin treatment are 2 major strategies for secondary stroke prevention.^{3,4} Dual antiplatelet therapy (DAPT) has been shown to be beneficial compared with aspirin alone in acute minor ischemic stroke or TIA.^{2,5,6} Recent studies showed that long-term intensive statin therapy was effective in reducing the risk of new stroke^{7,8} and early statin treatment was associated with better functional outcomes and lower mortality in patients with acute ischemic stroke or TIA.^{9,10}

The Intensive Statin and Antiplatelet Therapy for Acute High-risk Intracranial or Extracranial Atherosclerosis (INSPIRES) trial assessed the efficacy and safety of DAPT with clopidogrel-aspirin vs aspirin alone, immediate intensive statin vs 3-day delayed intensive statin, and the combination of DAPT and immediate intensive statin vs aspirin plus delayed intensive statin in patients with mild ischemic stroke or high-risk TIA of potentially atherosclerotic cause within 72 hours after symptom onset, using a factorial design.¹¹ Recent published results of the INSPIRES trial demonstrated that DAPT with clopidogrel-aspirin was superior to aspirin alone with a low but higher risk of moderate-to-severe bleeding, but no significant difference existed between immediate and delayed intensive statin treatment in reducing the risk of new stroke and increasing moderate-to-severe bleeding in these patients.^{12,13} However, the synergistic benefit of combining DAPT and immediate intensive statin therapy in patients with mild ischemic stroke or TIA within 72 hours after symptom onset is still unclear.

The primary research question addressed by this prespecified analysis in the INSPIRES trial was the interactive effects of DAPT with clopidogrel-aspirin and immediate intensive statin treatment in patients with mild ischemic stroke or high-risk TIA of presumed atherosclerotic cause within

72 hours after symptom onset, focusing on the efficacy and safety of the combination of DAPT with clopidogrel-aspirin and immediate intensive statin vs aspirin plus delayed intensive statin. The results of this analysis may provide important reference to the combined administration of intensive antiplatelet therapy and intensive statin in clinical practice.

Methods

Trial Design and Participants

The INSPIRES trial was a multicenter, randomized, double-blind, placebo-controlled trial with a 2-by-2 factorial design to assess the effect of clopidogrel-aspirin and immediate intensive statin on acute ischemic stroke or TIA. The study protocol and statistical analysis plan were published and are freely available as open access.¹¹ The INSPIRES trial enrolled 6,100 patients across 222 centers in China between September 17, 2018, and October 15, 2022. The trial enrolled patients aged 35–80 years who were diagnosed with mild ischemic stroke with a NIH Stroke Scale (NIHSS) score ≤ 5 or high-risk TIA (age, blood pressure, clinical features, duration of symptoms, and presence of diabetes [ABCD²] ≥ 4) within 24–72 hours after symptom onset or had an ischemic stroke with a NIHSS score 4–5 within 24 hours after symptom onset, with $\geq 50\%$ stenosis of a major intracranial or extracranial artery or acute multiple infarctions. Patients were ineligible to participate if they had received thrombolysis or endovascular therapy or had planned surgery or revascularization within 90 days; had planned nonstudy treatment with an additional anticoagulant, defibrinogenation, or antiplatelet therapy after the index event; had received DAPT with clopidogrel-aspirin or intensive statin therapy within 2 weeks before randomization; or had a history of intracranial hemorrhage, presumed cardioembolic TIA or ischemic stroke, or ischemic stroke or TIA of other determined etiologies (e.g., aortic dissection and vasculitis). Details of the inclusion and exclusion criteria for the trial are presented in eAppendix 1.

Randomization and Procedures

Eligible participants were randomly assigned within 72 hours after symptom onset, in a 1:1:1:1 ratio, to 1 of the 4 treatment groups: clopidogrel-aspirin plus immediate intensive statin, clopidogrel-aspirin plus delayed intensive statin, aspirin plus immediate intensive statin, and aspirin plus delayed intensive statin. Block randomization was performed using a block length of 8. Patients were randomly assigned a number corresponding to the study drug kit, and the medication in the kit was administered to the patient. The participants and the treating physicians were blinded to the treatment assignments.

Participants assigned to clopidogrel-aspirin received a 300-mg loading dose of clopidogrel on day 1 followed by 75 mg daily on days 2–90, plus aspirin at a dose of 100–300 mg on day 1 followed by 100 mg daily on days 2–21 and placebo aspirin on days 22–90. Participants taking aspirin alone received placebo clopidogrel for 90 days plus aspirin at a dose of 100–300 mg on day 1, followed by 100 mg daily on days 2–90. Participants who received immediate intensive statin therapy received 80 mg atorvastatin daily for days 1–21, followed by 40 mg atorvastatin daily on days 22–90. Participants assigned to delayed intensive statin therapy received placebo atorvastatin for days 1–3, followed by placebo atorvastatin and 40 mg atorvastatin daily on days 4–21 and 40-mg atorvastatin daily on days 22–90. Both atorvastatin 40 mg and 80 mg per day are currently recommended and clinically used, and atorvastatin at a dose of 40 mg daily may also achieve an intensive reduction in LDL-C levels similar to that achieved with 80 mg daily (47% vs 52%).¹⁴ Considering that the risk of bleeding might potentially increase when intensive statin, especially at higher doses, is combined with DAPT in the acute stage of ischemic stroke, a dose of 40 mg atorvastatin daily for days 4–21 was used in the delayed intensive statin group. After the 3-month study treatment, all patients received standard care based on the latest guidelines at the discretion of the local investigator and were followed up for an additional 9 months to assess clinical outcomes.

Outcomes

The primary efficacy outcome was a new stroke (ischemic or hemorrhagic) within 90 days. Secondary efficacy outcomes included a composite of vascular events (stroke, myocardial infarction, or vascular death) within 90 days, ischemic stroke, TIA, myocardial infarction, vascular death, and poor functional outcome (modified Rankin Scale [mRS] score of 2–6). A six-level ordered category scale that incorporated new stroke or TIA events and mRS score at 90 days was scored as follows: 5, fatal stroke (stroke with subsequent death); 4, severe stroke (stroke followed by mRS score 4–5); 3, moderate stroke (stroke followed by mRS score 2–3); 2, mild stroke (stroke followed by mRS score 0–1); 1, TIA; and 0, no stroke/TIA.¹⁵ Definitions of the clinical events are provided in eAppendix 2.

The primary safety outcome was moderate-to-severe bleeding events according to the Global Utilization of Streptokinase

and Tissue Plasminogen Activator for Occluded Coronary Arteries criteria at 90 days.¹⁶ The secondary safety outcomes included hepatotoxic effects, muscle toxic effects, all-cause mortality, intracranial hemorrhage, any bleeding, and additional adverse events or serious adverse events within 90 days.

All efficacy and safety outcomes were collected through face-to-face interviews with trained neurologists from the participating sites and reviewed and adjudicated by an independent clinical event adjudication committee.

Statistical Analysis

The risk of a new stroke over 90 days was presumed to be 13% in the group treated with aspirin alone plus delayed intensive statin therapy.^{17–19} We also assumed that both clopidogrel-aspirin therapy and immediate intensive statin treatment can reduce this risk by 22%,^{17,19} and the effects of intensive antiplatelet and lipid-lowering therapy will be similar and additive. With a total sample size of 6,100 participants (including 1,525 patients assigned to clopidogrel-aspirin plus immediate intensive statin and 1,525 patients assigned to aspirin alone plus delayed intensive statin), a two-sided α of 0.05, and 5% loss to follow-up, we calculated that this study would have 97% power to detect a conservative reduction in risk by 35% in the clopidogrel-aspirin plus immediate intensive statin group compared with the aspirin alone plus delayed intensive statin group.¹⁹

All analyses were performed in accordance with the intention-to-treat principle using the full analysis data set, which included all randomized patients. The test for a synergistic effect was preplanned and specified in the protocol and statistical analysis plan before unblinding. Interaction between antiplatelet and statin treatment assignments was evaluated by including antiplatelet treatment assignment, statin treatment assignment, and the interaction of antiplatelet-by-statin treatment assignment in the models. We estimated the cumulative risks of new clinical events within 90 days using Kaplan-Meier analysis. A Cox proportional hazards model with pooled trial centers ($n \geq 20$) set as a random effect was used to estimate treatment effects, and hazard ratios (HRs) and 95% CIs are presented. Patients who did not undergo a 90-day assessment were censored on the date of the event or the last visit if they withdrew from the trial or were lost to follow-up, whichever occurred first. The proportional hazards assumption was confirmed by assessing the interaction between the treatment assignment and the logarithmic function of survival time. Similar methods were used to compare the secondary efficacy outcomes of composite vascular events, ischemic stroke, hemorrhagic stroke, myocardial infarction, TIA, vascular death, and safety outcomes of bleeding and all-cause mortality. Shift analysis with ordinal logistic regression, for which the proportionality assumption was met, was used to compare the six-level scale of new stroke or TIA combined with the mRS, and common odds ratios (cORs) and 95% CIs were reported. Poor functional outcomes, hepatotoxicity, and muscle toxicity were compared using a logistic model, with

pooled trial centers set as a random effect, and ORs and 95% CIs were reported. In sensitivity analyses, we conducted similar analyses in the per-protocol population that completed the trial and had no major protocol deviations. Other adverse events and serious adverse events were compared using the χ^2 test or Fisher exact test. For the primary outcome, prespecified subgroup analyses were conducted, and interactions between treatment assignment and subgroups were evaluated in the models. In a post hoc analysis, we also performed a similar analysis for the outcome of net clinical benefit, defined as the composite of ischemic stroke, moderate-to-severe bleeding, and death.^{20,21} All data of 4 treatment groups were compared and included in the same model, and this report focused on presenting the effect of the combination of DAPT with clopidogrel-aspirin and immediate intensive statin vs aspirin plus delayed intensive statin.

All analyses were prespecified. No adjustment for multiple comparisons was planned, and analyses on secondary outcomes should be considered as exploratory. Two-sided *p* values < 0.05 were considered statistically significant. All analyses were performed using SAS software (version 9.4; SAS Institute, Cary, NC).

Standard Protocol Approvals, Registrations, and Patient Consents

The INSPIRES trial is registered at ClinicalTrials.gov (registration number: NCT03635749). The INSPIRES trial protocol was approved by the ethics committees of Beijing Tiantan Hospital (IRB approval number: KY2017-065-02) and all other participating sites. All participants or their representatives provided written informed consent before enrollment.

Data Availability

Researchers can request access to anonymized data for result reproduction or procedure replication by contacting the corresponding author. The study protocol and statistical analysis plan were published.

Results

From September 17, 2018, to October 15, 2022, 11431 patients with ischemic stroke or TIA were screened across 222 hospitals and 6,100 patients underwent randomization. Of them, 1,525 each were assigned to clopidogrel-aspirin plus immediate intensive statin, clopidogrel-aspirin plus delayed intensive statin, aspirin plus immediate intensive statin, and aspirin plus delayed intensive statin. At the 90-day follow-up, 3 patients in the clopidogrel-aspirin plus immediate intensive statin group and 1 patient in each of the other 3 groups were lost to follow-up and were treated as censored data (eFigure 1).

The baseline characteristics of the participants are summarized in Table 1 and eTable 1. The median age of the patients was 65 years, and 3,915 (64.2%) were male. Overall, 87.2% of patients were assigned to a treatment group within 24 hours

after symptom onset. Among them, 67.6% had multiple acute infarctions on imaging, 19.2% had single acute infarctions, and 13.1% had TIAs as index events. Among patients with acute ischemic stroke, 76.1% had NIHSS scores of 3 or less. Concomitant treatments during hospitalization and at 90-day follow-up are summarized in eTable 2.

There was no significant interaction between antiplatelet and statin treatment assignments in the 2 components of the factorial trial (*P* for interaction = 0.16 for the primary outcome). The primary outcome of new stroke within 90 days occurred in 116 patients (7.6%) in the clopidogrel-aspirin plus immediate intensive statin group (HR 0.76, 95% CI 0.60 to 0.97, *p* = 0.03; absolute risk reduction 2.2%, 95% CI 0.3%–4.1%; number needed to treat 45, 95% CI 24 to 302), in 106 patients (7.0%) in the clopidogrel-aspirin plus delayed intensive statin group (HR 0.69, 95% CI 0.54 to 0.89, *p* = 0.003), and in 129 patients (8.5%) in the aspirin plus immediate intensive statin group (HR 0.85, 95% CI 0.67 to 1.07, *p* = 0.17), compared with 150 patients (9.9%) in the aspirin plus delayed intensive statin group (Figure 1A and Table 2). Efficacy outcomes in the per-protocol population were consistent with those in the intention-to-treat population (eTable 3). The effect of treatment on the primary outcome was consistent across the prespecified subgroups (Figure 2 and eFigure 2). In a post hoc analysis, composite of ischemic stroke, moderate-to-severe bleeding, and death occurred in 124 patients (8.1%) in the clopidogrel-aspirin plus immediate intensive statin group (HR 0.77, 95% CI 0.61 to 0.97), in 121 patients (7.9%) in the clopidogrel-aspirin plus delayed intensive statin group (HR 0.75, 95% CI 0.59 to 0.95), and in 141 patients (9.2%) in the aspirin plus immediate intensive statin group (HR 0.88, 95% CI 0.70 to 1.11), compared with 158 patients (10.4%) in the aspirin plus delayed intensive statin group.

For secondary outcomes, composite vascular events occurred in 120 patients (7.9%) in the clopidogrel-aspirin plus immediate intensive statin group and in 151 patients (9.9%) in the aspirin plus delayed intensive statin group (HR, 0.78; 95% CI 0.61–0.99) (Table 3). The overall risks of ischemic stroke were 7.1% vs 9.7% (HR 0.72, 95% CI 0.56 to 0.92), favoring the clopidogrel-aspirin plus immediate intensive statin group. Poor functional outcome occurred in 143 of 1,523 patients (9.4%) in the clopidogrel-aspirin plus immediate intensive statin group (OR 0.72, 95% CI 0.57–0.90), in 158 of 1,524 patients (10.4%) in the clopidogrel-aspirin plus delayed intensive statin group (OR 0.81, 95% CI 0.64–1.01) and in 156 of 1,524 patients (10.2%) in the aspirin plus immediate intensive statin group (OR 0.79, 95% CI 0.63–0.99), compared with 190 of 1,522 patients (12.5%) in the aspirin plus delayed intensive statin group (Figure 3 and Table 2). There was no significant interaction between antiplatelet and statin treatment assignments for poor functional outcome (*P* for interaction = 0.49). A shift toward a less severe new stroke/TIA at 90 days was observed in the clopidogrel-aspirin plus immediate intensive statin group compared with the aspirin plus delayed intensive statin group

Table 1 Baseline Characteristics of the Participants in the INSPIRES Trial

Characteristics	Clopidogrel-aspirin plus immediate intensive statin (N = 1,525)	Clopidogrel-aspirin plus delayed intensive statin (N = 1,525)	Aspirin plus immediate intensive statin (N = 1,525)	Aspirin plus delayed intensive statin (N = 1,525)
Age, yr, median (IQR)	65 (57–71)	65 (57–71)	65 (57–71)	65 (56–71)
Female, no. (%)	495 (32.5)	568 (37.2)	561 (36.8)	561 (36.8)
Body mass index, kg/m ² , median (IQR)	24.5 (22.6–26.6)	24.2 (22.5–26.4)	24.5 (22.6–26.7)	24.5 (22.7–26.6)
Medical history, no. (%)				
Hypertension	1,024 (67.1)	1,023 (67.1)	1,020 (66.9)	1,016 (66.6)
Diabetes mellitus	404 (26.5)	426 (27.9)	409 (26.8)	419 (27.5)
Dyslipidemia	54 (3.5)	49 (3.2)	61 (4.0)	62 (4.1)
Previous ischemic stroke	468 (30.7)	433 (28.4)	459 (30.1)	449 (29.4)
Current smoking, no. (%)	439 (28.8)	453 (29.7)	437 (28.7)	454 (29.8)
Medicine use before qualifying events, no. (%)^a				
Aspirin	197 (12.9)	193 (12.7)	206 (13.5)	197 (12.9)
Clopidogrel	10 (0.7)	11 (0.7)	12 (0.8)	10 (0.7)
Lipid-lowering agents	149 (9.8)	147 (9.6)	153 (10.0)	138 (9.0)
Qualifying event, no. (%)				
TIA	208 (13.6)	191 (12.5)	221 (14.5)	181 (11.9)
Acute single infarction	285 (18.7)	303 (19.9)	298 (19.5)	288 (18.9)
Acute multiple infarctions	1,032 (67.7)	1,031 (67.6)	1,006 (66.0)	1,056 (69.2)
≥50% symptomatic stenosis, no. (%)^b				
Yes	1,215/1,491 (81.5)	1,233/1,494 (82.5)	1,231/1,488 (82.7)	1,236/1,495 (82.7)
No	276/1,491 (18.5)	261/1,494 (17.5)	257/1,488 (17.3)	259/1,495 (17.3)
Time to randomization after onset of symptoms, no. (%)				
≤24 h	207 (13.6)	194 (12.7)	184 (12.1)	198 (13.0)
>24–48 h	634 (41.6)	621 (40.7)	655 (43.0)	642 (42.1)
>48 h	684 (44.9)	710 (46.6)	686 (45.0)	685 (44.9)
NIHSS score in qualifying ischemic stroke, no. (%)				
≤3	996/1,317 (75.6)	1,011/1,334 (75.8)	1,004/1,304 (77.0)	1,022/1,344 (76.0)
>3	321/1,317 (24.4)	323/1,334 (24.2)	300/1,304 (23.0)	322/1,344 (24.0)
ABCD² score in qualifying TIA, no. (%)				
≤5	170/208 (81.7)	156/191 (81.7)	169/221 (76.5)	146/181 (80.7)
>5	38/208 (18.3)	35/191 (18.3)	52/221 (23.5)	35/181 (19.3)
Low-density lipoprotein cholesterol, mmol/L, median (IQR) ^c	2.49 (2.00–3.02)	2.62 (2.15–3.12)	2.49 (2.03–3.03)	2.59 (2.12–3.13)

Abbreviations: ABCD² = age, blood pressure, clinical features, duration of symptoms, and presence of diabetes; IQR = interquartile range; NIHSS = NIH Stroke Scale; TIA = transient ischemic attack.

^a Patients received medication within 1 mo before symptom onset.

^b Data were missing in 132 cases because of the absence of both intracranial and extracranial arterial vascular assessments, or participants did not have more than 50% stenosis in intracranial (or extracranial) arteries but extracranial (or intracranial) vascular assessments were missing.

^c Data were missing in 286 cases.

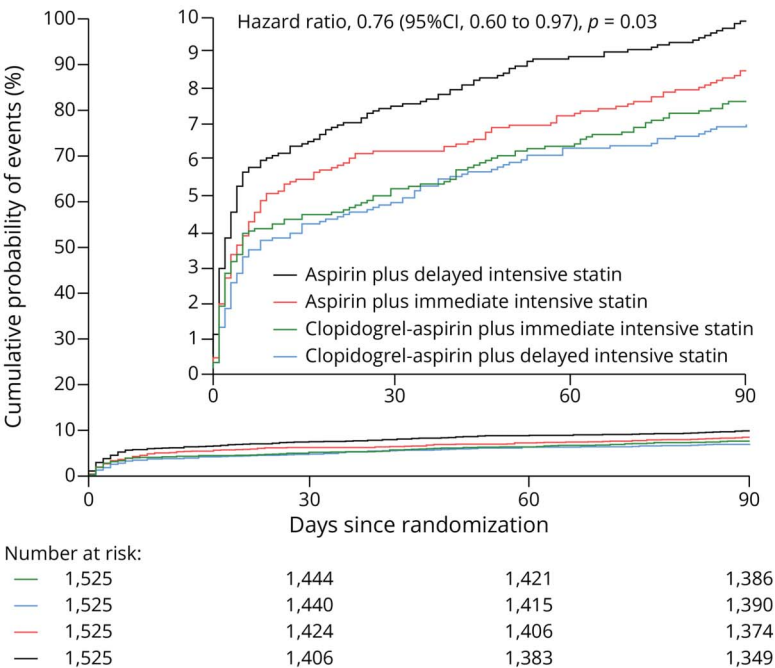
(cOR 0.73, 95% CI 0.57–0.93). Other secondary outcomes are presented in Table 3 and eTable 4.

There was no significant interaction between antiplatelet and statin treatment assignments in the risk of moderate-to-severe hemorrhage (*P* for interaction = 0.33 for the primary safety outcome). The primary safety outcome of moderate-to-severe

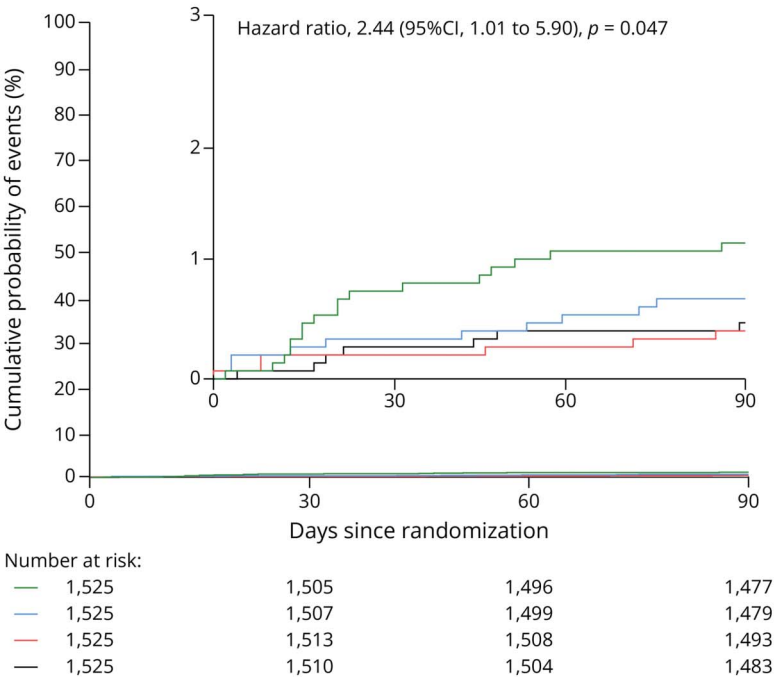
bleeding occurred in 17 patients (1.1%) in the clopidogrel-aspirin plus immediate intensive statin group (HR 2.44, 95% CI 1.01–5.90, *p* = 0.047), in 10 patients (0.7%) in the clopidogrel-aspirin plus delayed intensive statin group (HR 1.44, 95% CI 0.55–3.79, *p* = 0.46), and in 6 patients (0.4%) in the aspirin plus immediate intensive statin group (HR 0.87, 95% CI 0.29–2.58, *p* = 0.80), compared with 7 patients (0.5%)

Figure 1 Kaplan-Meier Curves for Stroke and Moderate-To-Severe Bleeding

A. New stroke



B. Moderate-to-severe bleeding



Cumulative Kaplan-Meier curves for new stroke (A) and moderate-to-severe bleeding (B). The hazard ratios, 95% CIs, and *p* values are presented for the comparison between combination treatment (clopidogrel-aspirin plus immediate intensive statin) and control group (aspirin plus delayed intensive statin).

Table 2 Primary Efficacy and Safety Outcomes

Outcomes	Antiplatelet treatment	Statin treatment		Hazard ratio/odds ratio ^b (95% CI)	<i>p</i> Value ^c
		Immediate intensive statin	Delayed intensive statin		
Primary efficacy outcome					
Stroke (ischemic or hemorrhagic)	Clopidogrel-aspirin	116/1,525 (7.6) ^a	106/1,525 (7.0)	1.10 (0.84–1.43)	0.49
	Aspirin	129/1,525 (8.5)	150/1,525 (9.9)	0.85 (0.67–1.07)	0.17
	Hazard ratio (95% CI)	0.89 (0.70–1.15)	0.69 (0.54–0.89)	0.76 (0.60–0.97) ^d	0.03
	<i>p</i> Value	0.38	0.003		
Secondary outcomes					
Poor functional outcome (mRS 2–6) ^e	Clopidogrel-aspirin	143/1,523 (9.4)	158/1,524 (10.4)	0.89 (0.70–1.13)	0.32
	Aspirin	156/1,524 (10.2)	190/1,522 (12.5)	0.79 (0.63–0.99)	0.04
	Odds ratio (95% CI)	0.91 (0.71–1.15)	0.81 (0.64–1.01)	0.72 (0.57–0.90) ^d	0.005
	<i>p</i> Value	0.43	0.06		
Primary safety outcomes					
Moderate-to-severe bleeding ^f	Clopidogrel-aspirin	17/1,525 (1.1)	10/1,525 (0.7)	1.70 (0.78–3.70)	0.19
	Aspirin	6/1,525 (0.4)	7/1,525 (0.5)	0.87 (0.29–2.58)	0.80
	Hazard ratio (95% CI)	2.82 (1.11–7.15)	1.44 (0.55–3.79)	2.44 (1.01–5.90) ^d	0.047
	<i>p</i> Value	0.03	0.46		

Abbreviations: mRS = modified Rankin scale; TIA = transient ischemic attack.

^a The event rates of poor functional outcomes (mRS score 2–6) were raw estimates, whereas the rates of stroke and bleeding were Kaplan-Meier estimates of the percentage of patients with events at 90 d.

^b Hazard ratios are shown for the stroke and bleeding outcomes and odds ratios for poor functional outcomes (mRS score 2–6).

^c No adjustment for multiple comparisons was planned and analyses on secondary outcomes should be considered as exploratory.

^d Hazard ratio or odds ratio for clopidogrel-aspirin plus immediate intensive statin vs aspirin plus delayed intensive statin.

^e mRS score data at 90 d were missing in 2 patients in the clopidogrel-aspirin plus immediate intensive statin group, 1 in the clopidogrel-aspirin plus delayed intensive statin group, 1 in the aspirin plus immediate intensive statin group, and 3 in the aspirin plus delayed intensive statin group.

^f Bleeding events were defined according to the Global Utilization of Streptokinase and Tissue Plasminogen Activator for Occluded Coronary Arteries criteria.

in the aspirin plus delayed intensive statin group (Figure 1B and Table 2). There were no significant differences between the clopidogrel-aspirin plus immediate intensive statin and the aspirin plus delayed intensive statin groups regarding hepatotoxic effects, muscle toxic effects, all-cause mortality, or any bleeding (Table 3). In the analyses of the per-protocol population, the safety outcomes were similar to those of the primary analyses (eTable 4). Adverse events occurred in 348 patients (22.8%) in the clopidogrel-aspirin plus immediate intensive statin group and 303 patients (19.9%) in the aspirin plus delayed intensive statin group (eTable 5); serious adverse events occurred in 54 patients (3.5%) and 47 patients (3.1%), respectively (eTable 6). Adverse events or serious adverse events leading to discontinuation of the trial treatment are listed in eTable 7.

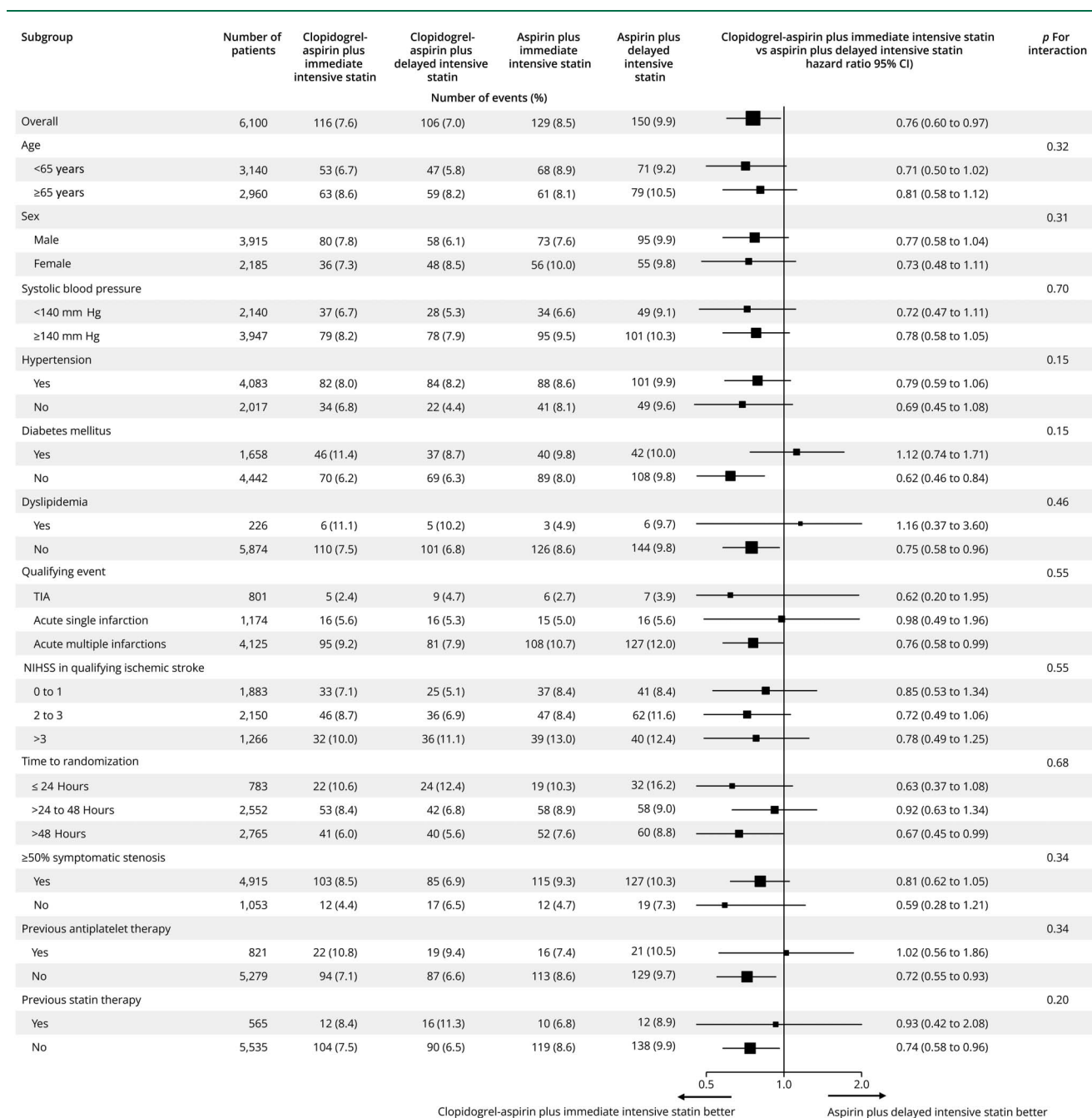
Classification of Evidence

This study provides Class I evidence that, in patients with mild ischemic stroke or TIA of presumed atherosclerotic cause, the combination of clopidogrel-aspirin plus delayed intensive statin was superior to aspirin plus delayed intensive statin in reducing the 90-day risk of new stroke.

Discussion

In the INSPIRES trial, which involved secondary stroke prevention of patients with mild ischemic stroke or TIA of presumed atherosclerotic cause within 72 hours after symptom onset, combination of DAPT with clopidogrel-aspirin and immediate intensive statin was associated with a 24% lower relative risk (2.2% lower absolute risk) of new stroke within 90 days compared with aspirin plus delayed intensive statin, which is mostly driven by clopidogrel-aspirin treatment, whereas, clopidogrel-aspirin alone seems to have a numerically lower risk of new stroke, although not significantly different from the combination treatment. Secondary analysis suggested that the combination treatment may be correlated with an improvement in functional outcomes at 90 days. Combination treatment demonstrated an approximately 2.4-fold higher risk of moderate-to-severe bleeding, which was also numerically higher than with either clopidogrel-aspirin treatment or immediate intensive statin treatment alone, although the risk was low across all groups. There was no significant increase in other adverse events such as hepatotoxic effects, muscle toxic effects, all-cause mortality, or any bleeding.

Figure 2 Hazard Ratios for the Stroke According to Prespecified Subgroups



The hazard ratios and their 95% CIs are presented for the comparison between combination treatment (clopidogrel-aspirin plus immediate intensive statin) and the control group (aspirin plus delayed intensive statin). Systolic blood pressure data were missing in 5 patients in the clopidogrel-aspirin plus immediate intensive statin group, 1 in the clopidogrel-aspirin plus delayed intensive statin group, 3 in the aspirin plus immediate intensive statin group, and 4 in the aspirin plus delayed intensive statin group. There were 34 patients in the clopidogrel-aspirin plus immediate intensive statin group, 31 in the clopidogrel-aspirin plus delayed intensive statin group, 37 in the aspirin plus immediate intensive statin group, and 30 in the aspirin plus delayed intensive statin group with missing data due to ≥50% symptomatic stenosis. NIHSS = NIH Stroke Scale; TIA = transient ischemic attack.

Antiplatelet therapy with clopidogrel and/or aspirin and intensive statins are frequently used for secondary stroke prevention in clinical practice.^{3,4} Previous study indicated that the effect of clopidogrel, which might be activated by *CYP3A4*, may be inhibited by atorvastatin, another *CYP3A4* substrate.²² However, a meta-analysis suggested that the effect of clopidogrel was mainly influenced by the *CYP2C19* variations,

rather than *CYP3A4*.²³ A previous study showed that atorvastatin reduced the ability of clopidogrel to inhibit platelet aggregation in a dose-dependent manner.²² The Fast Assessment of Stroke and TIA to prevent Early Recurrence trial (n = 392) implicated that early simvastatin use initiated within 24 hours after symptom onset may potentially attenuate the effect of DAPT with clopidogrel-aspirin in patients with

Table 3 Other Secondary Efficacy and Safety Outcomes

Outcomes	Clopidogrel-aspirin plus immediate intensive statin (N = 1,525), no. (%) ^a	Clopidogrel-aspirin plus delayed intensive statin (N = 1,525), no. (%) ^a	Aspirin plus immediate intensive statin (N = 1,525), no. (%) ^a	Aspirin plus delayed intensive statin (N = 1,525), no. (%) ^a	Clopidogrel-aspirin plus immediate intensive statin vs aspirin plus delayed intensive statin, hazard ratio/odds ratio ^b (95% CI)	p Value ^c
Secondary efficacy outcomes						
Composite vascular event (stroke, myocardial infarction, or vascular death)	120 (7.9)	109 (7.2)	131 (8.6)	151 (9.9)	0.78 (0.61–0.99)	0.04
Ischemic stroke	108 (7.1)	100 (6.6)	127 (8.3)	147 (9.7)	0.72 (0.56–0.92)	0.01
Recurrent stroke	79 (5.3)	80 (5.3)	94 (6.3)	111 (7.4)	0.70 (0.52–0.93)	0.01
TIA with infarction	3 (0.2)	2 (0.1)	7 (0.5)	4 (0.3)	0.73 (0.16–3.25)	0.68
Progressive stroke	26 (1.7)	18 (1.2)	26 (1.7)	32 (2.2)	0.81 (0.48–1.35)	0.41
Hemorrhagic stroke	9 (0.6)	6 (0.4)	2 (0.1)	3 (0.2)	3.00 (0.81–11.07)	0.10
TIA	8 (0.5)	13 (0.9)	21 (1.4)	18 (1.2)	0.44 (0.19–1.02)	0.06
Myocardial infarction	3 (0.2)	2 (0.1)	1 (0.1)	1 (0.1)	3.00 (0.31–28.85)	0.34
Vascular death	11 (0.7)	10 (0.7)	7 (0.5)	8 (0.5)	1.38 (0.55–3.42)	0.49
Ordinal stroke or TIA^d					0.73 (0.57–0.93)	0.01
Fatal stroke: mRS score 6	11/1,524 (0.7)	9/1,525 (0.6)	5/1,525 (0.3)	8/1,524 (0.5)		
Severe stroke: mRS scores 4–5	17/1,524 (1.1)	11/1,525 (0.7)	13/1,525 (0.9)	14/1,524 (0.9)		
Moderate stroke: mRS scores 2–3	30/1,524 (2.0)	39/1,525 (2.6)	41/1,525 (2.7)	61/1,524 (4.0)		
Mild stroke: mRS scores 0–1	57/1,524 (3.7)	47/1,525 (3.1)	70/1,525 (4.6)	66/1,524 (4.3)		
TIA	8/1,524 (0.5)	13/1,525 (0.9)	20/1,525 (1.3)	15/1,524 (1.0)		
No stroke or TIA	1,401/1,524 (91.9)	1,406/1,525 (92.2)	1,376/1,525 (90.2)	1,360/1,524 (89.2)		
Secondary safety outcomes						
Hepatotoxic effects^e	23 (1.5)	16 (1.0)	16 (1.0)	16 (1.0)	1.44 (0.76–2.75)	0.26
Muscle toxic effects^f	1 (0.1)	1 (0.1)	1 (0.1)	0 (0.0)	—	—
All-cause mortality	15 (1.0)	22 (1.4)	15 (1.0)	15 (1.0)	1.00 (0.49–2.04)	>0.99
Any bleeding	52 (3.4)	42 (2.8)	25 (1.6)	38 (2.5)	1.34 (0.88–2.04)	0.17
Intracranial hemorrhage	10 (0.7)	7 (0.5)	4 (0.3)	4 (0.3)	2.50 (0.78–7.98)	0.12
Mild bleeding^g	38 (2.5)	32 (2.1)	20 (1.3)	31 (2.0)	1.19 (0.74–1.91)	0.47

Abbreviations: mRS = modified Rankin scale; TIA = transient ischemic attack.

^a The event rates of ordinal stroke or TIA, hepatotoxic effects, and muscle toxic effects were raw estimates, whereas the rates of other outcomes were Kaplan-Meier estimates of the percentage of patients with events at 90 d.

^b Odds ratios are shown for hepatotoxic effects. The common odds ratios are shown for ordinal stroke or TIA. Hazard ratios are shown for the other outcomes.

^c No adjustment for multiple comparisons was planned, and analyses on secondary outcomes should be considered as exploratory.

^d The severity of stroke or TIA is classified on a six-level ordered categorical scale of combined vascular events with mRS. The mRS score at 90 d was missing in 1 patient each in the clopidogrel-aspirin plus immediate intensive statin group and the aspirin plus delayed intensive statin group.

^e Hepatotoxic effects were defined as alkaline phosphatase or aspartate aminotransferase levels >3 times the upper limit of the normal value.

^f Muscle toxic effects were defined as creatine kinase >10 times the upper limit of the normal value or the presence of muscle pain, myopathy, or rhabdomyolysis.

^g Bleeding events were defined according to the Global Utilization of Streptokinase and Tissue Plasminogen Activator for Occluded Coronary Arteries criteria.

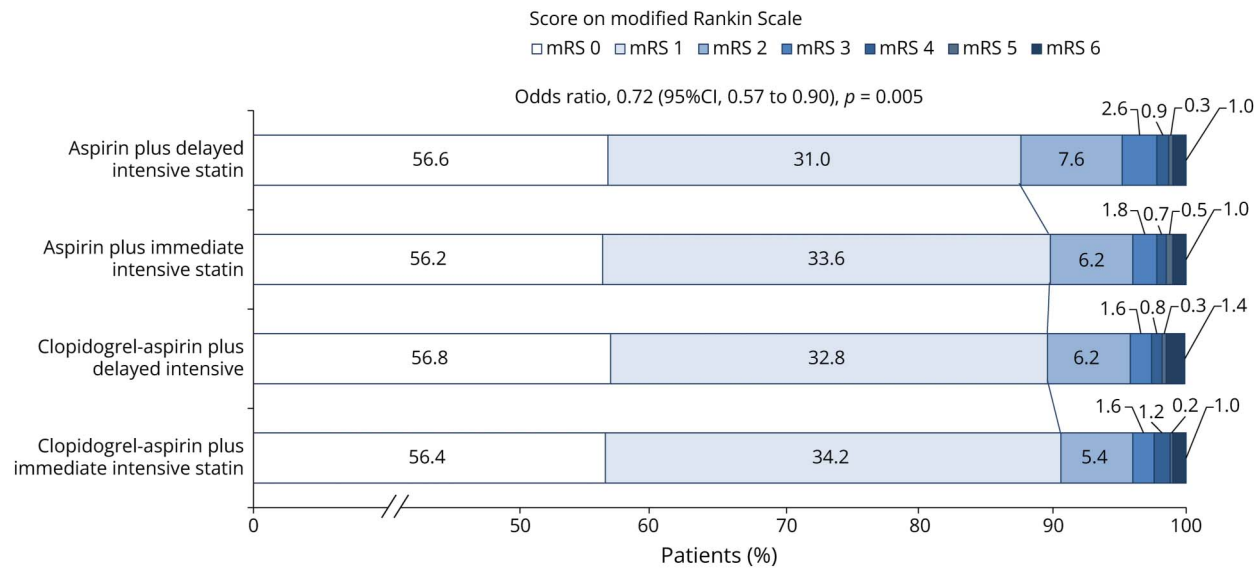
minor stroke or TIA.²⁴ This analysis of the INSPIRES trial suggested that early atorvastatin did not significantly attenuate the effect of clopidogrel-aspirin (event rate 7.6% vs 7.0% in clopidogrel-aspirin with and without immediate intensive statin) and there is no synergistic effect of clopidogrel-aspirin and immediate intensive statin on reduction of risk of new stroke. In INSPIRES, combination of DAPT and immediate intensive statin initiated within 72 hours may reduce the risk of new stroke. However, our study showed that there is a possible synergistic effect on increasing the risk of hemorrhage and combining DAPT with immediate intensive statin therapy in the acute phase may increase bleeding risk compared with aspirin plus delayed intensive statin and clopidogrel-aspirin plus delayed intensive statin, although the overall moderate-to-severe risk was low (1.1% vs 0.5% vs 0.7%) and potentially caused by chance. This analysis of the INSPIRES trial suggested that patients within 72 hours of mild ischemic stroke or TIA of presumed atherosclerotic cause can benefit from the combination of clopidogrel-aspirin and immediate intensive statin in reducing risk of new stroke, although there was more moderate-to-severe bleeding that does not seem to offset this benefit. However, clopidogrel-aspirin plus delayed intensive statin seems to be associated with a numerically lower risk of new stroke and lower risk of hemorrhage. Nevertheless, it should be acknowledged that the benefit or harm effect of statin treatment in this trial may be explained by difference in not only the timing of initiation of statin (immediate vs 3-day delayed) but also the intensity of statin treatment (80 mg vs 40 mg atorvastatin daily for days 4–21).

Furthermore, combination treatment was associated with an improvement in functional outcomes at 90 days, although it is

hypothesis-generating for secondary outcomes. The magnitude of reduction in new stroke for combination treatment was somewhat smaller than that for clopidogrel-aspirin plus delayed intensive statin (absolute risk reduction 2.2% vs 2.9%), whereas the magnitude of improvement of functional outcome was larger (absolute risk reduction 3.1% vs 2.1%). This implies that, in addition to the reduction in new stroke, the extra benefit of combination treatment in improving functional outcome might be partially achieved because of the neuroprotective effects of immediate statin administration. In addition to lowering low-density lipoprotein cholesterol, statins have multifaceted neuroprotective effects, such as enhancing endothelial function, reducing inflammation, and having antiplatelet and antithrombotic effects.²⁵ Our study provides evidence of the benefits of the combination of clopidogrel-aspirin and immediate intensive statin for acute ischemic stroke patients but needs further validation, especially for functional outcome as a primary outcome.

This study has a few limitations. First, patients with presumed cardioembolic causes, those with moderate or severe stroke, and candidates for thrombolysis or thrombectomy were excluded from the trial; thus, the results cannot be generalized to these populations. Second, clopidogrel is a prodrug that requires hepatic cytochrome P450 (CYP), enzymes to generate active metabolites²⁶; however, the CYP genotype relevant to drug metabolism was not included in the trial entry criteria. Third, the trial exclusively enrolled Chinese patients, who may be more susceptible to bleeding,²⁷ have a high prevalence of intracranial artery stenosis (30%–50% in Asians vs 8% in non-Asians),²⁸ and exhibit genetic differences in hepatic enzymes and drug transporters^{26,29}; hence, the

Figure 3 Distribution of the Modified Rankin Scale Score at 90 Days



The odds ratio and its 95% CI are presented for the comparison between combination treatment (clopidogrel-aspirin plus immediate intensive statin) and the control group (aspirin plus delayed intensive statin). mRS = modified Rankin Scale.

external generalizability of these findings to White and Black populations needs further validation.

In conclusion, in patients with mild ischemic stroke or TIA of presumed atherosclerotic cause within 72 hours after symptom onset, those who received clopidogrel-aspirin and delayed intensive statin had a lower risk of new stroke within 90 days, mainly driven by clopidogrel-aspirin and not significantly different from that with clopidogrel-aspirin plus immediate intensive statin, but possibly a synergistic effect of clopidogrel-aspirin and immediate intensive statin on increasing the risk of hemorrhage. Furthermore, combination treatment might be related to improved functional outcomes at 90 days compared with aspirin plus delayed intensive statin treatment, which requires further validation.

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Author Contributions

Y. Pan: drafting/revision of the manuscript for content, including medical writing for content; study concept or design. Y. Gao: major role in the acquisition of data. W. Chen: major role in the acquisition of data. S.C. Johnston: critical revision of the manuscript for content. P. Amarenco: critical revision of the manuscript for content. P.M. Bath: critical revision of the manuscript for content. J. Jing: major role in the acquisition of data. C. Wang: major role in the acquisition of data. L. Jiang: major role in the acquisition of data. H. Yan: analysis or interpretation of data. X. Wang: analysis or interpretation of data. Y. Yang: major role in the acquisition of data. T. Wang: major role in the acquisition of data. X. Meng: major role in the acquisition of data. J. Lin: major role in the acquisition of data. X. Zhao: major role in the acquisition of data. L. Liu: major role in the acquisition of data. Y. Wang: major role in the acquisition of data. Y. Wang: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design.

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